

Constrained optimisation and KKT

Minimise $f(x)$ subject to $h_i(x) = 0$ and $g_j(x) \leq 0$: Lagrange multipliers, slack and tight inequalities, the four KKT conditions, a worked example and the dual.

THE QUESTION

Setting $\nabla f = 0$ ignores the constraints. How do we find the best point when x must stay on a curve or inside a region?

THE INTUITION

At a constrained minimum no allowed move goes downhill, so a contour of f touches the constraint and the gradients line up. An inequality is either slack, with multiplier 0, or tight, with a positive one.

KEY TAKEAWAY

Form $\mathcal{L} = f + \sum_i \lambda_i h_i + \sum_j \mu_j g_j$. Then solve $\nabla_x \mathcal{L} = 0$, $h_i = 0$, $g_j \leq 0$, $\mu_j \geq 0$ and $\mu_j g_j = 0$, one guess of tight constraints at a time.

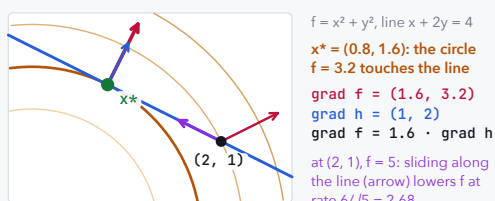
min f , $h = 0$, $g \leq 0 \rightarrow \mathcal{L} = f + \lambda h + \mu g \rightarrow \nabla_x \mathcal{L} = 0 \rightarrow \mu \geq 0, \mu g = 0 \rightarrow \text{try cases} \rightarrow x^*$

1 A constraint moves the minimum off $\nabla f = 0$



Minimise $f(x)$ over $x \in \mathbb{R}^k$ subject to m equalities $h_i(x) = 0$ and n inequalities $g_j(x) \leq 0$; write $x \geq 0$ as $-x \leq 0$. When the unconstrained minimiser breaks a constraint, the answer lies on the constraint and $\nabla f \neq 0$ there.

2 On a line, the best point is where a contour touches it



The lecture's example with only its equality: minimise $x^2 + y^2$ subject to $h = x + 2y - 4 = 0$. If ∇f had a component along the line, sliding along it would lower f . So at the answer ∇f is normal to the line, like ∇h (gradients cross level sets at right angles; the maths for ML sheet): $\nabla f = -\lambda \nabla h$.

3 Lagrange multipliers turn the picture into equations

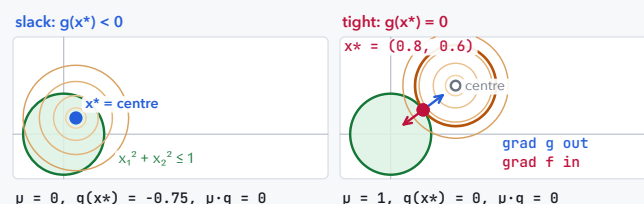


Set every partial derivative of $\mathcal{L} = x^2 + y^2 + \lambda(x + 2y - 4)$ to 0:

$$2x + \lambda = 0, \quad 2y + 2\lambda = 0, \quad x + 2y = 4.$$

The first two are $\nabla f + \lambda \nabla h = 0$; the third, $\partial \mathcal{L} / \partial \lambda = 0$, is the constraint. So $x = 0.8$, $y = 1.6$, $\lambda = -1.6$, $f = 3.2$.

4 An inequality is either slack or tight



Minimise $\|x - c\|^2$ subject to $g = x_1^2 + x_2^2 - 1 \leq 0$ (ours). With c inside the disc the constraint is slack: $x^* = c$, $g < 0$, $\mu = 0$. With $c = (1.6, 1.2)$ outside it is tight: $x^* = c / \|c\| = (0.8, 0.6)$, $g = 0$ and $\mu = \|c\| - 1 = 1$. Either way $\mu_j g_j(x^*) = 0$: complementary slackness.

5 The multiplier of an inequality is never negative

At a tight constraint ∇g points out of the region, where g grows. Stationarity gives $\nabla f = -\mu \nabla g$. With $\mu \geq 0$, f rises into the region and every downhill step leaves it, so no feasible step improves; above, $\nabla f = (-1.6, -1.2) = -1 \cdot \nabla g$. With $\mu < 0$ a step inward would lower f . An equality blocks both sides, so its λ has either sign. The slides write $\mu_i = \nabla_x f / \nabla_x \mu_i g_i$; the argument needs $\nabla f = -\mu_i \nabla g_i$.

6 The four KKT conditions

Step 1: $\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_{i=1}^m \lambda_i h_i(x) + \sum_{j=1}^n \mu_j g_j(x)$. At a constrained minimum (with well-behaved constraints) all four hold:

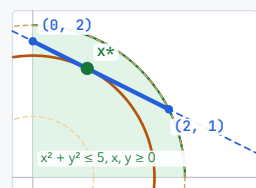
- Stationarity** $\nabla f + \sum_i \lambda_i \nabla h_i + \sum_j \mu_j \nabla g_j = 0$ (step 2)
- Primal feasibility** $h_i(x) = 0$ (step 3) and $g_j(x) \leq 0$
- Dual feasibility** $\mu_j \geq 0$; each λ_i any sign
- Compl. slackness** $\mu_j g_j(x) = 0$ for every j (step 4)

The slides' summary leaves out $g_j(x) \leq 0$; check it too. For a convex problem, such as the ridge budget or the SVM, a point that passes all four is the global minimum.

The lecture's example, case by case, and the dual

Minimise $x^2 + y^2$ subject to $x^2 + y^2 \leq 5$, $x + 2y = 4$ and $x, y \geq 0$; then lower bounds and duality.

7 Write the example as f , one h and three g 's



the lecture's example
feasible set: the thick segment
from (2, 1) to (0, 2)

f is 5 at (2, 1), 4 at (0, 2) and
dips to 3.2 at $x^* = (0.8, 1.6)$

the circle $f = 3.2$ touches the line
inside the disc and the quadrant,
so g_1 and g_3 are all slack

$h = x + 2y - 4$, $g_1 = x^2 + y^2 - 5$, $g_2 = -x$, $g_3 = -y$, so
 $\mathcal{L} = x^2 + y^2 + \lambda(x + 2y - 4) + \mu_1(x^2 + y^2 - 5) - \mu_2x - \mu_3y$.

Stationarity: (1) $2x + \lambda + 2\mu_1x - \mu_2 = 0$, (2) $2y + 2\lambda + 2\mu_1y - \mu_3 = 0$. **Constraint:** (3) $x + 2y = 4$. **Slackness:** (4) $\mu_1(x^2 + y^2 - 5) = 0$, (5) $\mu_2x = 0$, (6) $\mu_3y = 0$.

8 Try every pattern of tight constraints

active g	point	multipliers	verdict
none	(0.8, 1.6)	$\lambda = -1.6$	all four hold
circle	(2, 1)	$\lambda = 0, \mu_1 = -1$	$\mu_1 < 0$
circle	(-0.4, 2.2)	$\lambda = 0, \mu_1 = -1$	$x < 0, \mu_1 < 0$
$x = 0$	(0, 2)	$\lambda = -2, \mu_2 = -2$	$\mu_2 < 0$
$y = 0$	(4, 0)	$\lambda = -8, \mu_3 = -16$	outside disc, $\mu_3 < 0$

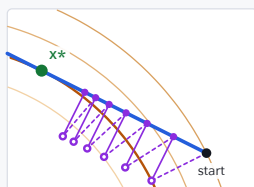
any two or three no point: (0, 2) and (4, 0) are not on the circle

(4) to (6) each split two ways: 8 patterns. With two or more g_j tight there is no point. Only "nothing tight" passes: (0.8, 1.6), $\lambda = -1.6$, $\mu = 0$. The slides get this answer, but $y = 0$ fails because (4, 0) is outside the disc and needs $\mu_3 = -16$, not because of (1), which holds with $\lambda = -8$. $x = 0$ fails because it needs $\mu_2 = -2$.

9 Check all four, then check by brute force

stationarity	$\text{grad } f + \lambda \text{ grad } h = (0, 0)$
primal: h	$0.8 + 2 \cdot 1.6 - 4 = 0$
primal: g	$g = (-1.8, -0.8, -1.6)$, all < 0
dual	$\mu = (0, 0, 0) \geq 0$, $\lambda = -1.6$ free
slackness	$\mu_j g_j = 0$ for all three

projected gradient, $\alpha = 0.1$, from (2, 1):
step, project back to the segment;
within $1e-6$ of x^* after 64 steps



At (0.8, 1.6), $\nabla f = (1.6, 3.2)$ cancels $\lambda \nabla h = -1.6 \cdot (1, 2)$, and all three g_j are slack, so $\mu = 0$ and every $\mu_j g_j = 0$. A grid of 450,001 points on the line keeps the feasible ones ($1 \leq y \leq 2$) and finds the minimum 3.2 at $y = 1.6$. Projected gradient descent takes a step, projects back onto the feasible segment and repeats; from (2, 1) it is within 10^{-6} of the answer after 64 steps.

10 The Lagrangian is a lower bound



$L(x, \mu) = (x - 2)^2 + \mu(x - 1)$
black: $J(x)$, a wall at $x = 1$
dots: $D(\mu)$, the min over x

$\mu = 0$ $D = 0$
 $\mu = 1$ $D = 0.75$
 $\mu = 2$ $D = 1$
 $\mu = 4$ $D = 0$

every $L \leq J$ where $x \leq 1$;
best: $\mu = 2$ gives $1 = p^*$

The notes' exact penalty: $J(x) = f(x) + \sum_i 1(g_i(x))$, where $1(z)$ is 0 for $z \leq 0$ and ∞ otherwise. Exact, but as hard as before. For $\mu_i \geq 0$ and feasible x , each $\mu_i g_i \leq 0$, so $\mathcal{L}(x, \mu) \leq J(x)$, and $J(x) = \max_{\mu \geq 0} \mathcal{L}$. The problem is $\min_x \max_{\mu \geq 0} \mathcal{L}$. The dual function $D(\mu) = \min_x \mathcal{L}(x, \mu)$ is the lowest point of each curve; here (ours) $D(\mu) = \mu - \mu^2/4 \leq p^* = 1$. The notes write λ for these multipliers.

11 Weak duality always; strong for SVM-like problems

$q = xy$ (the notes)

1	1	2	3	4
2	2	4	6	8
3	3	6	9	12
4	4	8	12	16

x 1 2 3 4 y

max min = 4 = min max
ours: $[[0,1], [1,0]]$ gives $0 < 1$



The minimax inequality $\max_y \min_x q \leq \min_x \max_y q$ (the notes' table $q = xy$ gives 4 and 4) applied to $\mathcal{L}$ gives weak duality: $p^* = \min_x \max_{\mu \geq 0} \mathcal{L} \geq \max_{\mu \geq 0} D(\mu) = d^*$. For the example,

$$D(\lambda, \mu) = -\frac{(\lambda - \mu_2)^2 + (2\lambda - \mu_3)^2}{4(1 + \mu_1)} - 4\lambda - 5\mu_1.$$

At $\mu = 0$ it is $-1.25\lambda^2 - 4\lambda$, largest at $\lambda = -1.6$: $d^* = 3.2 = p^*$. No random multiplier (200,000 tried) beats 3.2. SVM-like problems have $d^* = p^*$ (strong duality); others can keep a duality gap $p^* - d^* > 0$.

12 Pause and think

$\mu_j > 0$ while $g_j < 0$? Never: slackness forces one of them to 0.

Ridge budget $\theta^T \theta \leq S$? $\lambda = 0$ if least squares already fits inside the ball, else the constraint is tight (the ridge regression sheet).

Which training points matter in an SVM? Those with a multiplier > 0 , on the margin (next sheet).

What does a QP solver take? $\min \frac{1}{2} x^T P x + q^T x$ subject to $Gx \leq h$, $Ax = b$.

$$\mathcal{L} = f + \sum \lambda_i h_i + \sum \mu_j g_j \cdot \nabla_x \mathcal{L} = 0 \cdot h_i = 0, g_j \leq 0 \cdot \mu_j \geq 0 \cdot \mu_j g_j = 0 \cdot d^* \leq p^*$$

Before: the reinforcement learning sheet; next: the support vector machines I (maximum margin) sheet. Source: lecture slides "KKT conditions" and "Constrained optimization" I and II (the notes "Lagrangian and duality"), and the SVM with cvxopt notebook for the QP form (nipunbatra.github.io/ml-teaching). The one-variable problem, the unit-disc cases and the random multipliers are ours, seeded. Recomputed: the slides' reasons for ruling out $y = 0$ and $x = 0$.